Elementary Charge and Millikan Experiment

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Introduction

The purpose of this experiment is to determine the elementary charge e using charged oil droplets in the Millikan experiment. From their rise and fall times at different voltages, we calculate the droplets' charges and radii, then determine e from their velocities in and against the direction of gravity. Finally, we estimate the electron's e/m ratio.

Theory and Methods

The rising and falling motion of charged oil droplets in the electric field of the capacitor is observed.

The force F applied to a sphere of radius r and velocity v in a viscous liquid of viscosity η is given by Stokes' law:

$$F_d = 6\pi r\eta v \tag{1}$$

A droplet of mass m, volume V, and density ρ_1 is located in Earth's gravitational field.

$$F_g = mg = \rho_1 V g \tag{2}$$

Buoyant force F_b is given by:

$$F_b = \rho_2 V g, \quad \rho_2 \text{ :is the density of air} \tag{3}$$

Electric force F_e is given by:

$$F_e = QE, \quad E = \frac{U}{d} \tag{4}$$

While Q is the charge on the droplets, ($Q = ne$) U is the applied voltage, and d is the distance between the plates of the capacitor.

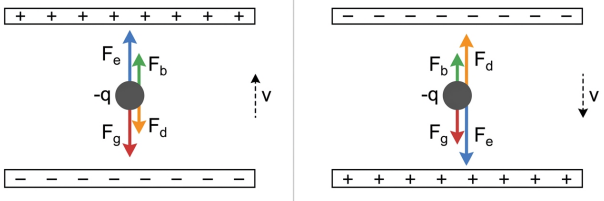


Figure: Forces acting on a droplet in the Millikan experiment.

1. Falling Droplet (v_1)
When the droplet is falling, the forces acting on it are:

$$F_b + F_d = F_g + F_e \tag{5}$$

$$\frac{4}{3}\pi r^3\rho_2 g + 6\pi\eta r v_1 = \frac{4}{3}\pi r^3\rho_1 g + Q\frac{U}{d} \tag{6}$$

$$6\pi\eta r v_1 = Q\frac{U}{d} + \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2) \tag{7}$$

$$v_1 = \frac{1}{6\pi\eta r}\left(Q\frac{U}{d} + \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{8}$$

2. Rising Droplet (v_2)
When the droplet is rising, the forces acting on it are:

$$F_b + F_e = F_g + F_d \tag{9}$$

$$\frac{4}{3}\pi r^3\rho_2 g + Q\frac{U}{d} = \frac{4}{3}\pi r^3\rho_1 g + 6\pi\eta r v_2 \tag{10}$$

$$6\pi\eta r v_2 = Q\frac{U}{d} - \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2) \tag{11}$$

$$v_2 = \frac{1}{6\pi\eta r}\left(Q\frac{U}{d} - \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{12}$$

By subtracting the equation for v_2 from v_1, the electric force terms cancel out, allowing us to isolate the radius r:

$$v_1 - v_2 = \frac{4r^2 g(\rho_1 - \rho_2)}{9\eta}$$

$$r^2 = \frac{9\eta(v_1 - v_2)}{4g(\rho_1 - \rho_2)}$$

$$r = \frac{3}{2}\sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}}\sqrt{|v_1 - v_2|}$$

By grouping the experimental constants into C_2, the equation for the radius simplifies to:

$$C_2 = 6.37 \cdot 10^{-5}\,(\mathrm{m}\cdot\mathrm{s})^{1/2} \tag{13}$$

$$r = C_2\sqrt{|v_1 - v_2|} \tag{14}$$

Similarly, by adding the equations for v_1 and v_2, the gravitational and buoyant force terms cancel out, allowing us to find the charge Q:

$$v_1 + v_2 = \frac{1}{6\pi\eta r}\left(2Q\frac{U}{d}\right) = \frac{QU}{3\pi\eta r d}$$

$$Q = \frac{3\pi\eta d(v_1 + v_2)r}{U}$$

Substituting the full expression for r derived above into this equation gives:

$$Q = \left(\frac{9}{2}\pi d\sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}}\right)\frac{v_1 + v_2}{U}\sqrt{|v_1 - v_2|}$$

By grouping these constants into C_1, the final equation

for the charge of the droplet is obtained:

$$C_1 = 2.73 \cdot 10^{-11}\,\mathrm{kg \cdot m(m \cdot s)^{-1/2}} \tag{15}$$

$$Q = C_1 \frac{v_1 + v_2}{U}\sqrt{|v_1 - v_2|} \tag{16}$$

Experimental Procedure

The eyepiece micrometer was calibrated using a stage micrometer. Oil droplets were introduced with the sprayer, and the microscope was focused on a charged droplet. Using the commutator switch, the droplet was moved between two calibrated positions while its ascent and descent times were recorded. The measurements were repeated at 400 V and 500 V. The distances and times were then used to calculate the velocities, charges, and radii of the droplets.

Results

Table 1: Measured ascent and descent data

U (V)	t_1 (s)	s_1 (div)	s_1 (mm)	t_2 (s)	s_2 (div)	s_2 (mm)
400	3.23	30	0.89	3.25	30	0.89
400	3.43	30	0.89	3.06	30	0.89
400	1.65	30	0.89	2.33	30	0.89
500	3.23	30	0.89	3.53	30	0.89
500	2.12	30	0.89	2.82	30	0.89

Table 2: Calculated velocities, charge, and radius

U (V)	v_1 falling (m/s)	v_2 rising (m/s)	Q (C)	r (m)
400	2.7×10^{-4}	2.8×10^{-4}	4.9×10^{-20}	8.3×10^{-8}
400	2.9×10^{-4}	2.6×10^{-4}	2.1×10^{-19}	3.6×10^{-7}
400	3.8×10^{-4}	5.4×10^{-4}	7.9×10^{-19}	8.0×10^{-7}
500	2.5×10^{-4}	2.8×10^{-4}	1.4×10^{-19}	3.1×10^{-7}
500	3.2×10^{-4}	4.2×10^{-4}	4.1×10^{-19}	6.5×10^{-7}
		Average Q:	3.2×10^{-19}	–

Sample calculations: For the first 400 V measurement,

$$v_1 = \frac{0.00089}{3.25} = 2.73846 \times 10^{-4}\ \mathrm{m/s},$$

$$v_2 = \frac{0.00089}{3.23} = 2.75542 \times 10^{-4}\ \mathrm{m/s}.$$

Using the manual constants $C_1 = 2.73 \times 10^{-11}$ and $C_2 = 6.37 \times 10^{-5}$,

$$\begin{aligned} Q &= (2.73 \times 10^{-11})\frac{(2.73846 \times 10^{-4}) + (2.75542 \times 10^{-4})}{400} \\ &\quad \times \sqrt{|2.73846 \times 10^{-4} - 2.75542 \times 10^{-4}|} \\ &= 4.8826 \times 10^{-20}\ \mathrm{C}, \\ r &= (6.37 \times 10^{-5})\sqrt{|2.73846 \times 10^{-4} - 2.75542 \times 10^{-4}|} \\ &= 8.2948 \times 10^{-8}\ \mathrm{m}. \end{aligned}$$

For the first 500 V measurement, $v_1 = 0.00089/3.53 = 2.5 \times 10^{-4}$ m/s and $v_2 = 0.00089/3.23 = 2.8 \times 10^{-4}$ m/s. Therefore, $Q = 1.4 \times 10^{-19}$ C and $r = 3.1 \times 10^{-7}$ m.

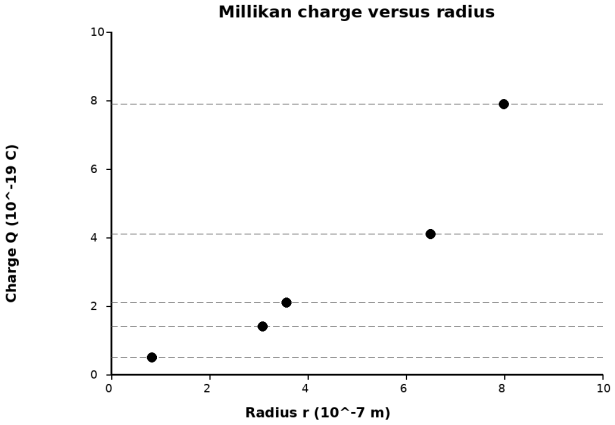


Figure: Measured charge Q as a function of droplet radius r.

Table 3: Results

Investigated levels	ΔQ (C)
$Q = 0$ to first level	4.9×10^{-20}
First to second level	9.1×10^{-20}
Second to third level	7.0×10^{-20}
Third to fourth level	2.0×10^{-19}
Fourth to fifth level	3.8×10^{-19}
Average	1.580×10^{-19}

The elementary charge was estimated from the average charge-level difference:

$$e_{\mathrm{exp}} = 1.580 \times 10^{-19}\ \mathrm{C}.$$

This is close to the literature value $e_{\mathrm{lit}} = 1.602 \times 10^{-19}$ C.

Discussion

The measured elementary charge is close to the accepted value. The percentage error is

$$\frac{|1.580 - 1.602|}{1.602} \times 100 = 1.37\%.$$

The small difference may come from errors in measuring time and distance, focusing the microscope, and reading the micrometer. The graph shows different measured charge values. Their average spacing gives an estimate of the elementary charge.

Only five droplets were measured, so more measurements would give a stronger result. The electron mass was not measured; therefore, e/m_e was calculated using the accepted mass:

$$\frac{e}{m_e} = \frac{1.580 \times 10^{-19}}{9.109 \times 10^{-31}} = 1.734 \times 10^{11}\ \mathrm{C/kg}.$$

Overall, the measurements at 400 V and 500 V gave $e = 1.580 \times 10^{-19}$ C and $e/m_e = 1.734 \times 10^{11}$ C/kg. The e/m_e value is derived because the electron mass was taken from the accepted value.